

Vouch SCORED26: Authenticated-Native Artifact Boundary Implementation and Reproduction Record

Artifact release record

Abstract

This record reports the implemented authenticated-native boundary for Vouch. It separates structural acceptance from authenticated origin, binds Native evidence to a consumer-supplied trust policy, and promotes evidence to a decision only after transcript agreement and release signability checks. The reported fixture, workload, mutation, and performance values are rendered from digest-verified owner reports after the signed release chain is checked. The result is evidence about observed agreement over an explicit checked surface; it is not a proof of policy correctness or whole-language semantic equivalence.

1 Problem and threat model

Native-shaped JSON is not evidence of native origin. A verifier must first apply bounded canonical parsing, select an already-authorized Ed25519 key from an out-of-band consumer policy, verify the DSSE signature, and only then interpret the authenticated payload. Bridge evidence remains a distinct artifact class and cannot be promoted through the Native decision boundary.

The threat model includes artifact-class confusion, context substitution, payload/envelope mismatch, a path replaced after command entry, a source inode mutated after its digest check, cross-release mixing, and a caller attempting to reach a release key with an unsigned or non-signable result. The model does not assume an honest application after a capability has been returned. It also excludes freshness, signer compromise, dishonest input collection, legal fitness, and the correctness of the policy itself.

2 Mechanism and implementation

The checked lane evaluates the same normalized source and immutable input in a reference evaluator and a separately implemented Meaning Environment evaluator. Invocation-bound, one-shot trace tokens prevent a caller from assembling a receipt from serialized transcripts. Issuance requires complete terminals, byte-identical observable traces, an empty diagnostic set, a final decision value, a release build, and a non-mutant build before the key handle is resolved. Refusals before key resolution are tested with a counting key provider and perform zero key operations.

Verification uses a fixed order: resource and canonical gates; closed policy and envelope schemas; selected-key authorization; signature verification; payload canonicity and schema; intrinsic receipt consistency; and expected context comparison. Successful verification yields authenticated evidence, not a decision. A separate promotion step checks agreement, completed terminals, trace completeness, the final decision, and release metadata.

The release lifecycle is acyclic. A signed deterministic descriptor D binds the source commit, archive, executable, toolchains, dependency manifests, and exact-reproduction results. A trusted Linux outer driver authenticates D before extracting an immutable private snapshot, executes phase 1, and constructs the clean-run report Q . Phase 2 derives and signs observation R from D , Q , and the exact owner-report buffers; publication record P names D and R . Phase 3 verifies $D \rightarrow Q \rightarrow R \rightarrow P$, replays the final derivations, verifies the paper-source snapshot, renders this document, and emits terminal report S .

3 Implementation validation

The phase-1 fixture report records 163 matched of 163 built fixtures, with 0 mismatches and 0 skips. It records 0 remaining design-target rows. These values are observations from the verified fixture report, not constants embedded in this paper.

The frozen workload contains 1536 candidates and 240 selected cases. There are 240 comparable decision pairs, 83 decision flips, and 19 held-out flips. These results describe only the supplied deterministic workload.

The mutation campaign seeded 12 and built 12 compile-time mutants. The frozen workload activated 5; 4 were detected, for a mutant-level detection rate of 33.3%. Detection is an empirical diagnostic, not a conformance threshold.

The clean run took 3942 seconds. The trusted outer driver recorded 482 deterministic exact-reproduction comparisons, while 4 owner reports remained verify-only observations.

Table 1: Five warmups followed by thirty measured runs; nearest-rank statistics.

Metric	Statistic	Value	Unit	Population
envelope_bytes	maximum	49214	byte	480
envelope_bytes	median	49194	byte	480
envelope_bytes	p95	49210	byte	480
native_verification_latency	maximum	1313897	microsecond	14400
native_verification_latency	median	27347	microsecond	14400
native_verification_latency	p95	68042	microsecond	14400
peak_resident_memory	maximum	74588160	byte	30
peak_resident_memory	median	74440704	byte	30
peak_resident_memory	p95	74588160	byte	30
selected_corpus_replay_latency	maximum	47888411	microsecond	30
selected_corpus_replay_latency	median	7602479	microsecond	30
selected_corpus_replay_latency	p95	22105416	microsecond	30

4 Release identity

The following values are injected only after the publication checker verifies the descriptor and signed observation. They are absent from the source commit and distributed archive as resolved publication fields.

Source commit:

44cd740d8615cfefb82b047520bcd5111d49d19

Freeze ancestor:

24b703f6b3ecc100d38a642d2b2ca1257717abb6

Release key identifier:

sha256:dfad0e0c05811e9c83c5733eaf2e5009a9bf11c8877278400116d66d774bd982

Release descriptor digest:

sha256:d7a3dccbd78d1ecbfe89093a949cee3f1567ae0c288535c2178b6b3ff14fbbb3

Reproduction observation digest:

sha256:b62a0b3e4f6e2baf4e8153ee56d3c1a9d8bd256bb44c6e85c0c0738b82c6b5f3

Archive digest:

sha256:c65d79d8a1deb2230e841793142410472f0aa7b23ef3809ce893b65edf513aa6

Engine digest:

sha256:ec176101243830abb860e152c444487ad0c61566d20351a8dadd98851cb57a04

Target:

x86_64-unknown-linux-gnu

5 Limitations

The two evaluators share numeric and value substrates, so common-mode errors remain possible. A receipt establishes observed agreement for the recorded source, input, profile, engine, budgets, and transcript boundary. It does not establish semantic equivalence for Lispex in general, independent witnessing, freshness, fairness, legality, generation honesty outside the signed release boundary, or protection against a dishonest consumer that ignores the API. Unsalted public hashes do not protect low-entropy real-world records. Public welfare-shaped inputs in this artifact are synthetic and generated by the committed workload procedure; they are not drawn from people, case files, production logs, or administrative datasets.

6 Artifact availability

The release set consists of the source archive, signed descriptor, clean-run report, signed reproduction observation, publication record, resolved paper, and terminal publication report. Authorization comes from the consumer's out-of-band trust policy; the adjacent public-key record and unsigned publication record are distribution and indexing aids, not trust roots.